University of Glasgow

Data fusion and extremes: a match made in Bayesian heaven

Daniela Castro-Camilo

Joint work with D. Cuba, C. Wilkie and M. Scott

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Key question

How can we fuse data with a particular focus at improving modelling and prediction of extremes?

- Statistical data fusion is a highly active area of research, often appearing under different names such as data melding, calibration, or certain forms of downscaling
- Here we will focus on the problem of fusing two data sources that aim to inform about the same process (PM2.5 levels), but are measured in different ways. In particular, in different spatio-temporal domains.

Lit Review on Bayesian DF

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- Bayesian hierarchical models provide a natural framework for data fusion

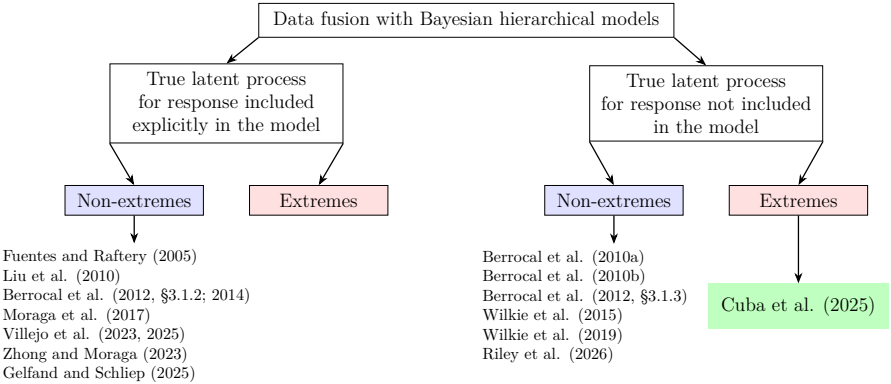
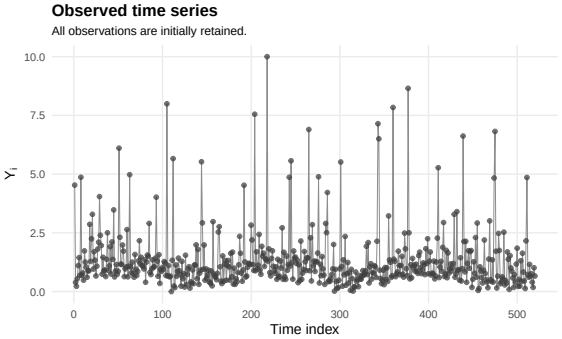


Figure 1: *Figure co-created with my PhD student Teng Wei Yeo.*

What do we mean by extremes?



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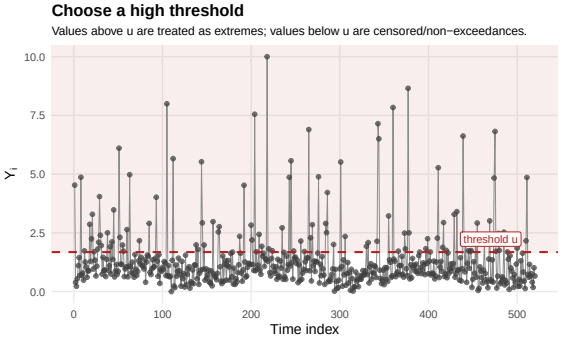

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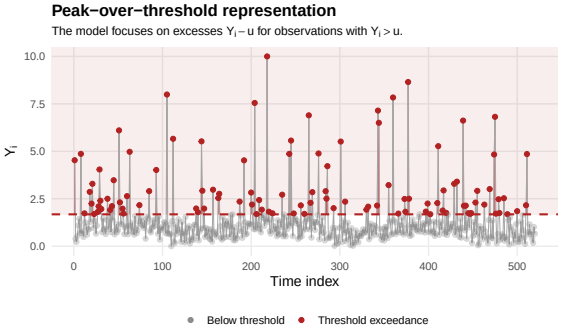

Under mild conditions, $Y - u \mid Y > u \sim H$, where H is the generalised Pareto distribution,

$$H(y \mid \sigma, \xi) = 1 - \left(1 + \frac{\xi y}{\sigma}\right)^{-1/\xi}, \qquad y > 0.$$

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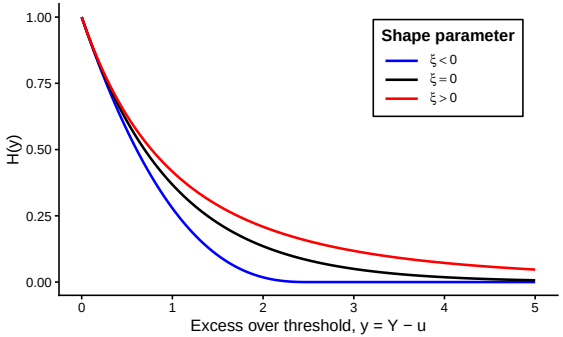

Special cases: Uniform ($\xi = -1$), Exponential ($\xi = 0$), Pareto-like ($\xi > 0$)

Our setup & goal & approach in one slide

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Improve the representation of extreme values in remote-sensing data using in-situ measurements and threshold-based statistics of extremes.

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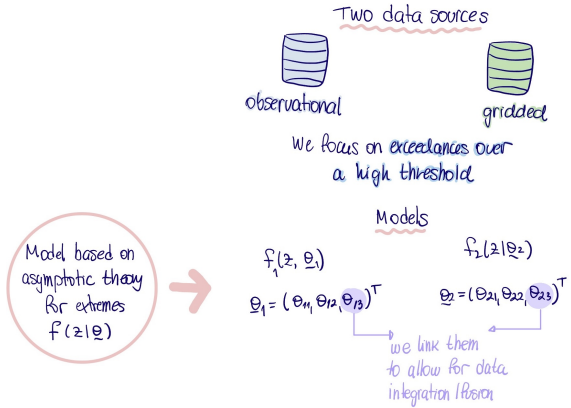
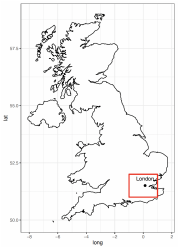


Air quality monitoring ($PM_{2.5}$) in the Great London area

- PM concentrations in the UK are available through the Automatic Urban and Rural Network (AURN).
- Network is extensive, but spatial coverage is insufficient to inform targeted local policies required for the effective improvement of air quality.
- Temporal coverage of the stations is not uniform and data have missing obs.



More about the data

Two sources of information:
AURN (in-situ) and CAMSRA (ECMWF Atmospheric Composition Reanalysis 4). We focus on the year 2022.

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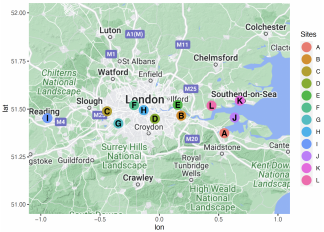
AURN

- Most stations register main air pollutants at hourly intervals.
- In the Great London region, there are 26 active stations, but only 12 contain records for more than 75% of the days in 2022.
- Air pollution guidelines are on a 24-hour average scale ⇒ we aggregate to daily means.



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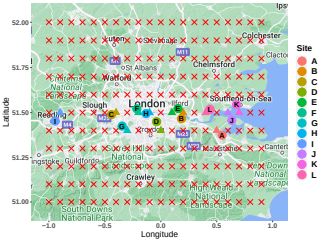
CAMSRA

- Global reanalysis dataset of atmospheric composition.
- Aggregated from the sub-daily scale to the daily average.
- 220 available cells (at a spatial scale of $0.1° \times 0.1°$)
- Choose the grid cells that contain the AURN stations.



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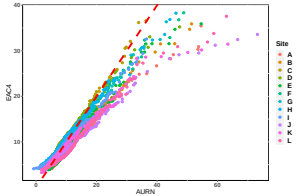
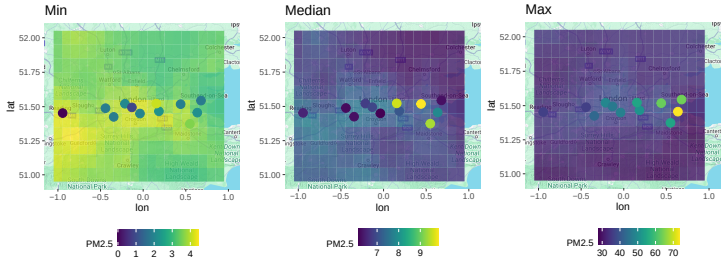


Top: Comparing data from each AURN site with the corresponding nearest grid centroid from the CAMSRA dataset.

Bottom: Minimum, median and maximum of the CAMSRA data and AURN observation stations.



Extreme Data Fusion (exDF) approach

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$$f(z|p,\sigma,\xi) = (1-p_u)\delta(z) + \frac{p_u}{\sigma_u}\left(1+\frac{\xi z}{\sigma_u}\right)^{-1/\xi-1}\Delta_0(z)$$

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$$\text{logit}\left(p_{u,y_{ij}^*}\right) = \lambda_{i0} + \lambda_{i1}\mathbb{1}(x_{ij-1}^*>0) + \lambda_{i2}\mathbb{1}(x_{ij}^*>0) + \lambda_{i3}\mathbb{1}(x_{ij+1}^*>0),$$

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- Exc prob for $\mathbf{x}$ (gridded): $p_x = \mathbb{1}(x^*>0)$

$$f(z|p,\sigma,\xi) = (1-p_u)\delta(z) + \frac{p_u}{\sigma_u}\left(1+\frac{\xi z}{\sigma_u}\right)^{-1/\xi-1}\Delta_0(z)$$

- ξ_y and ξ_x: constant across space and time. Constraints are imposed to preserve the first and second moments of the δ-GPD and ensure the regularity of maximum likelihood estimators (MLEs). This can be achieved with a scaled Laplace prior on $(-0.5, 0.5)$, which shrinks ξ to 0 (in the spirit of PC priors).
- $\sigma_{u_y,i}$ and $\sigma_{u_x,i}$: scale params warrant special attention, as they encapsulate spatio-temporal information and play a key role in our data fusion approach. We model them as smooth temporal curves, fitted using basis functions of the same dimension. Specifically, we assume that

$$\log\{\sigma_{u_y,i}\} = \Phi_i \boldsymbol{c}_i, \qquad \log\{\sigma_{u_x,i}\} = \Psi_i \boldsymbol{d}_i, \quad i = 1,\ldots,n,$$

where Φ_i and Ψ_i are matrices of basis functions of dimensions $j_i \times m$ and $k_i \times m$, respectively. Recall, $j_i \neq k_i \Rightarrow$different temporal supports are allowed.

The full model is

$$
\begin{aligned}
\boldsymbol{y}_i^* \mid \boldsymbol{c}_i &\sim \delta\text{-GPD}\left(\exp(\Phi_i \boldsymbol{c}_i), \xi_y, \mathbf{p}_{yi}\right), \\
\text{logit}\left(p_{u,y_{ij}}\right) &= \lambda_{i0} + \lambda_{i1}\mathbb{1}(x^*_{ij-1} > 0) + \lambda_{i2}\mathbb{1}(x^*_{ij} > 0) + \lambda_{i3}\mathbb{1}(x^*_{ij+1} > 0) \\
c_{ir} \mid \alpha_{ir}, \beta_{ir}, d_{ir}, \sigma_c^2 &\sim \text{N}\left(\alpha_{ir} + \beta_{ir} d_{ir}, \sigma_c^2\right), \quad r = 1, \ldots, m, \\
\boldsymbol{x}_i^* \mid \boldsymbol{d}_i &\sim \delta\text{-GPD}\left(\exp(\Psi_i \boldsymbol{d}_i), \xi_x, \mathbf{p}_{xi}\right), \\
p_{u,x_{ik}} &= \mathbb{1}(x^*_{ik} > 0), \\
\boldsymbol{d}_i &\sim \text{N}_m\left(\boldsymbol{\mu}_d, \Sigma_d\right), \\
\xi_y &\sim \text{Laplace}_{\xi}(\mu_y, b_y) \quad \text{where } -0.5 < \xi_y < 0.5, \\
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\boldsymbol{\alpha}_j = (\alpha_{1r}, \ldots, \alpha_{nr})^\top \mid \sigma_\alpha^2 &\sim \text{N}_n\left(\mathbf{0}, \sigma_\alpha^2 \exp\left(-\phi_\alpha \Sigma_{\text{data}}\right)\right), \\
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\left(\sigma_\alpha^2\right)^{-1} &\sim \mathrm{Ga}\left(a_\alpha, b_\alpha\right), \\
\left(\sigma_\beta^2\right)^{-1} &\sim \mathrm{Ga}\left(a_\beta, b_\beta\right), \\
\left(\sigma_c^2\right)^{-1} &\sim \mathrm{Ga}\left(a_c, b_c\right), \\
\lambda_{il} &\sim N(\mu_{\lambda_l}, \sigma^2_{\lambda_l}), \quad l = 0, 1, 2, 3,
\end{aligned}
$$

The full model is

$$
\begin{aligned}
\boldsymbol{y}_i^* \mid \boldsymbol{c}_i &\sim \delta\text{-GPD}\left(\exp(\Phi_i \boldsymbol{c}_i), \xi_y, \mathbf{p}_{yi}\right), \\
\text{logit}\left(p_{u,y_{ij}}\right) &= \lambda_{i0} + \lambda_{i1}\mathbb{1}(x^*_{ij-1} > 0) + \lambda_{i2}\mathbb{1}(x^*_{ij} > 0) + \lambda_{i3}\mathbb{1}(x^*_{ij+1} > 0) \\
c_{ir} \mid \alpha_{ir}, \beta_{ir}, d_{ir}, \sigma_c^2 &\sim \text{N}\left(\alpha_{ir} + \beta_{ir} d_{ir}, \sigma_c^2\right), \quad r = 1, \ldots, m, \\
\boldsymbol{x}_i^* \mid \boldsymbol{d}_i &\sim \delta\text{-GPD}\left(\exp(\Psi_i \boldsymbol{d}_i), \xi_x, \mathbf{p}_{xi}\right), \\
p_{u,x_{ik}} &= \mathbb{1}(x^*_{ik} > 0), \\
\boldsymbol{d}_i &\sim \text{N}_m\left(\boldsymbol{\mu}_d, \Sigma_d\right), \\
\xi_y &\sim \text{Laplace}_\xi(\mu_y, b_y) \quad \text{where } -0.5 < \xi_y < 0.5, \\
\xi_x &\sim \text{Laplace}_\xi(\mu_x, b_x), \quad \text{where } -0.5 < \xi_x < 0.5. \\
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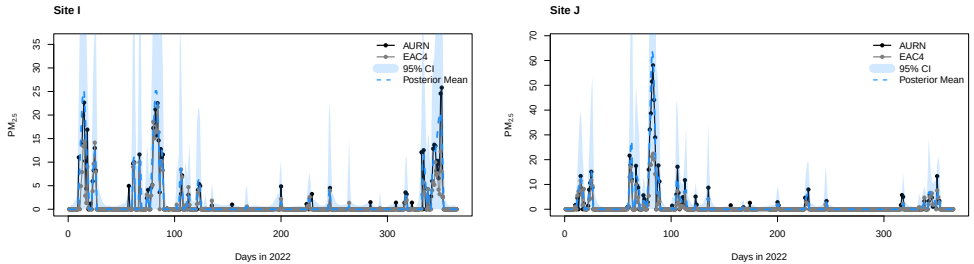
\end{aligned}
$$

- Many choices made. The most important ones are:
  - **Thresholds** u_x, u_y: We use graphical diagnostics to select the threshold for the AURN data (MRL plots: theoretical mean excess function is linear in the threshold), identifying the 80th station-wise empirical quantile as suitable, retaining about 75 observations per station. We use the same quantile for the CAMSRA data.
  - **Decay params** ϕ_α **and** ϕ_β: pragmatic approach, assumed fixed after estimated outside the model.
  - **Basis dimension** set to $m = 60$ via LOSO-CV.
- Implementation: MCMC in C++.

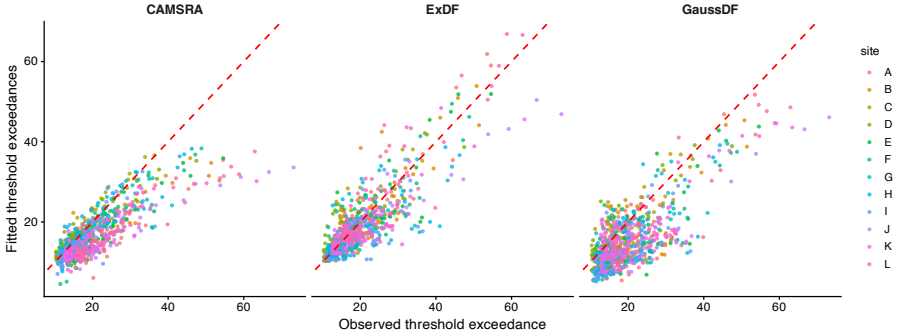
Example of model fit: $PM_{2.5}$ measurements from sites I (West) and J (East).



- Posterior means closely align with the AURN data, effectively capturing both threshold exceedances ($\boldsymbol{y}_i > 0$) and censored non-exceedances ($\boldsymbol{y}_i = 0$).
- 95% credible intervals are relatively wide.
- $\hat{\xi}_y = -0.10[-0.216, 0.05]$ and $\hat{\xi}_x = -0.22[-0.34, 0.02]$. Pointwise estimates suggest bounded tails (finite upper limits), but accounting for uncertainty indicates that unbounded tails remain plausible.

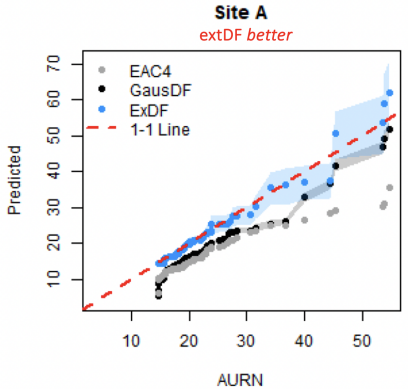
Comparison across all locations: CAMSRA, ExDF, GaussDF

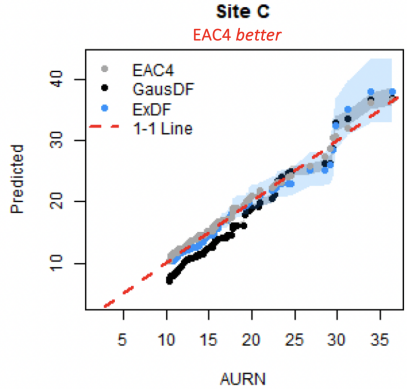
Observed AURN exceedances with corresponding CAMSRA exceedances and fitted exceedances (posterior means) from the ExDF and GaussDF (appied to log-transformed measurements) models across all 12 monitoring sites.



Model validation

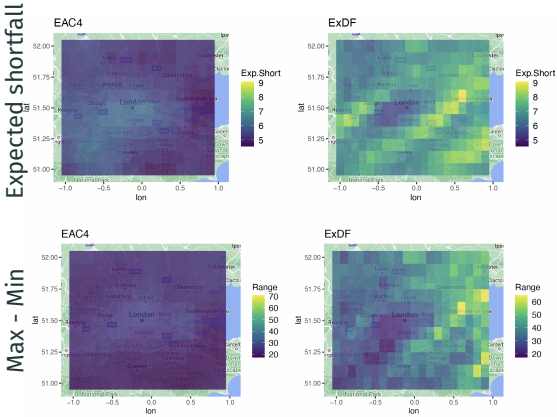
- Leave-one-site-out CV at the 12 stations.
- exDF vs CAMSRA: consistently better in 10/12 sites (RMSE/MAE).
- exDF vs GaussDF: exDF always better (RMSE/MAE/CRPS)





CAMSRA vs exDF: Visual comparison of exceedances

- Figure shows expected shortfall (ES, average of exceedances) and range of exceedances (max-min).



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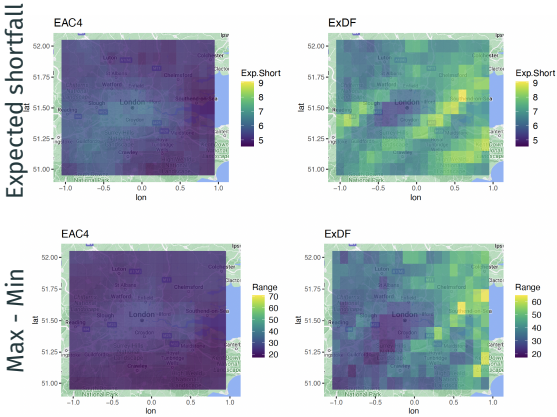

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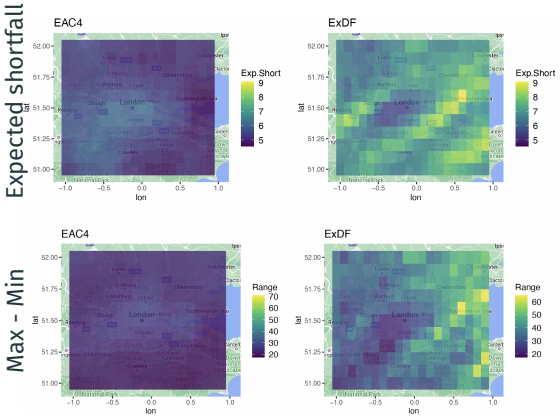

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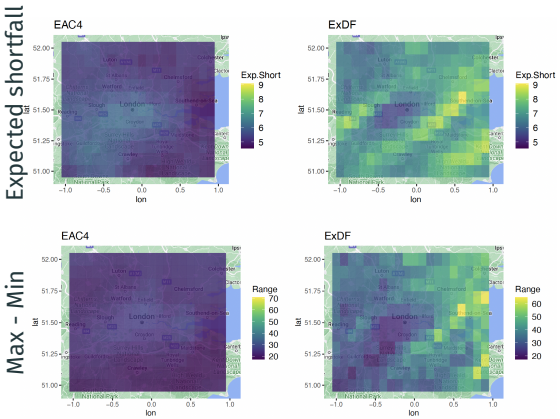
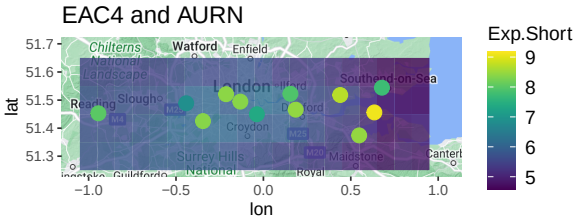


- ES and range for CAMSRA are smooth in space with little variability.
- Less visible patterns seem to agree (northwest, city centre to the southeast)
- Main differences: CAMSRA shows higher values near the London city centre. ExDF data shows higher exceedances on the East Coast, consistent with the literature (Yang et al., 2023).


EAC4 and AURN
lat
lon
51.7
51.6
51.5
51.4
51.3
−1.0
−0.5
0.0
0.5
1.0
Exp.Short
9
8
7
6
5
Chilterns
National
Landscape
Watford
Enfield
London
Southend-on-Sea
Reading
Slough
Croydon
Surrey Hills
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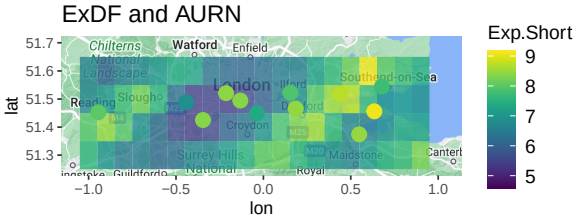
M4
M2
M25
M20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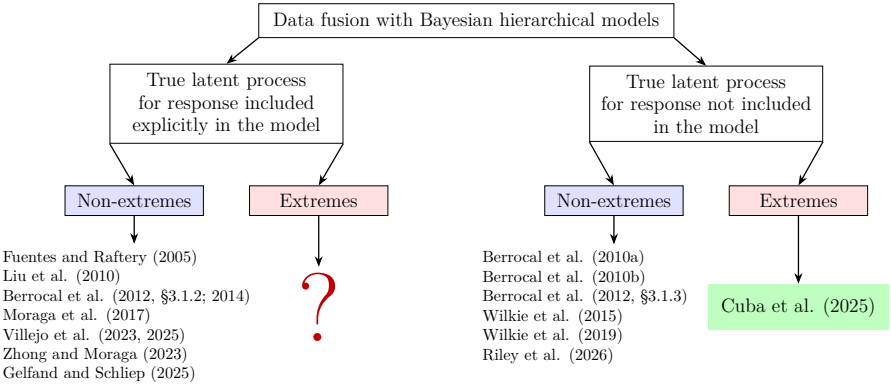
- Integrated estimation of exponential decay params.

- Including the true latent/unobserved process explicitly into the modelling



Charla basada en:

Cuba et al. (2026). **Spatio-temporal fusion of reanalysis and in situ data for censored threshold exceedances of $PM_{2.5}$.** *The Annals of Applied Statistics (to appear).*

Accede al paper

¡Gracias!

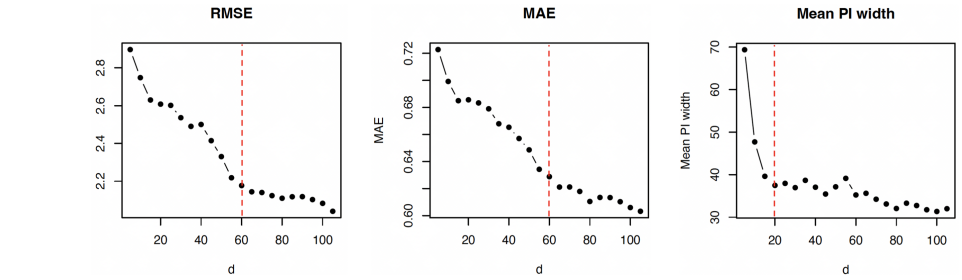
¿Preguntas?

References

- Berrocal, V. J., Gelfand, A. E. and Holland, D. M. (2010a). A Bivariate Space–Time Downscaler under Space and Time Misalignment. *AoAS* 4.4, pp. 1942–1975.
- Berrocal, V. J., Gelfand, A. E. and Holland, D. M. (2010b). A Spatio-Temporal Downscaler for Output From Numerical Models. *JABES* 15.2, pp. 176–197.
- Fuentes, M. and Raftery, A. E. (2005). Model Evaluation and Spatial Interpolation by Bayesian Combination of Observations with Outputs from Numerical Models. *Biometrics* 61.1, pp. 36–45.
- Gelfand, A. E., Schliep, E. M. (2025). Model-Based Spatial Data Fusion. *Annual Review of Statistics and Its Application*, 13.
- Riley, A. I., Blangiardo, M., Piel, F. B., Beddows, A., Beevers, S., Fuller, G. W., Agnew, P. and Pirani, M. (2026). A Bayesian Multisource Fusion Model for Spatiotemporal PM2.5 in an Urban Setting. *Environmetrics* 37.1, e70065
- Villejo, S. J., Illian, J. B. and Swallow, B. (2023). Data Fusion in a Two-Stage Spatio-Temporal Model Using the INLA-SPDE Approach. *Spatial Statistics* 54, p. 100744.
- Villejo, S. J., Martino, S., Lindgren, F. and Illian, J. B. (2025). A Data Fusion Model for Meteorological Data Using the INLA-SPDE Method. *JRSS-C* 74.4, pp. 1021–1056.
- Wilkie, C. J., Miller, C. A., Scott, E. M., O'Donnell, R. A., Hunter, P. D., Spyrakos, E. and Tyler, A. N. (2019). Nonparametric Statistical Downscaling for the Fusion of Data of Different Spatiotemporal Support. *Environmetrics* 30.3, e2549.
- Wilkie, C. J., Scott, E. M., Miller, C., Tyler, A. N., Hunter, P. D. and Spyrakos, E. (2015). Data Fusion of Remote-sensing and In-lake Chlorophylla Data Using Statistical Downscaling. *Procedia Environmental Sciences. Spatial Statistics Conference* 2015 26, pp. 123–126.

Choice of dimension for basis functions

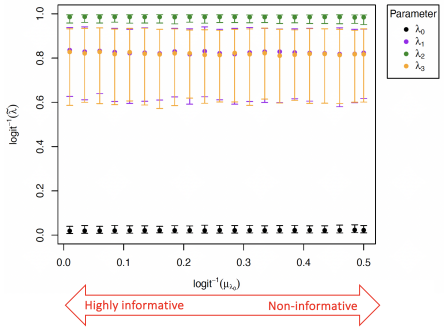
- Sensitivity analysis using LOO.
- Plots shows RMSE/MAE/Mean PI (avg width of 95% predictive intervals) width averaged across all locations.



- Increasing d improves the model and increases computation costs.
- Elbow plots for RMSE, MAE, and mean PI show that beyond $d = 60$, improvements are marginal.

Sensitivity on logistic regression coefficients

- Study to assess the sensitivity of the model to the prior specifications of the λ parameters.
- Figure shows changes in the posterior given different prior specifications from highly informative to uninformative.



- Posterior are robust, converge to similar values.

Table 6.1: Mean(95% credible interval) of $\hat{\lambda}_0$, $\hat{\lambda}_1$, $\hat{\lambda}_2$ and $\hat{\lambda}_3$ averaged over the 11 fitted sites when predicting over site L.

Parameter	Estimate	Probability
$\hat{\lambda}_0$	-2.95(-3.82,-2.07)	0.05(0.02,0.11)
$\hat{\lambda}_1$	0.85(-0.53,2.23)	0.7(0.37,0.90)
$\hat{\lambda}_2$	2.19(0.81,3.58)	0.9(0.69,0.97)
$\hat{\lambda}_3$	0.85(-0.55,2.23)	0.7(0.37,0.90)

- on average, $\text{logit}^{-1}(\hat{\lambda}_0) = 0.05$, meaning that the model assigns a 0.05 probability of exceeding the threshold to all times points, irrespective of the presence of a threshold exceedance in the CAMSRA dataset.
- $\hat{\lambda}_2$ (linked to time t), indicates there is a 0.9 probability of observing an exceedance in y given an exceedance in x.
- $\hat{\lambda}_1$ and $\hat{\lambda}_3$ (linked to times $t-1$ and $t+1$) have confidence intervals inclusive of 0.

Table 6.2: Classification metrics for the logistic regression component of the ExDF model for fitted sites A to K. Table includes a comparison with the EAC4 data.

Site	Accuracy	Precision	Recall	Specificity	F1 Score
A	0.94	0.84	0.89	0.96	0.86
B	0.96	0.87	0.95	0.97	0.91
C	0.96	0.87	0.92	0.97	0.89
D	0.97	0.90	0.95	0.97	0.92
E	0.91	0.79	0.76	0.95	0.77
F	0.94	0.83	0.89	0.95	0.86
G	0.95	0.85	0.92	0.96	0.88
H	0.92	0.77	0.85	0.93	0.81
I	0.90	0.75	0.79	0.93	0.77
J	0.95	0.84	0.92	0.96	0.88
K	0.95	0.86	0.92	0.96	0.89
EAC4	0.95	0.87	0.87	0.97	0.87